

Basic Principles of Medicine 2

Module: Digestion and Metabolism

Lecture: DM 29 Heme Synthesis and Porphyrrias

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Acknowledgement: Dr. Margit Trotz, Ph.D.
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Recommended Reading

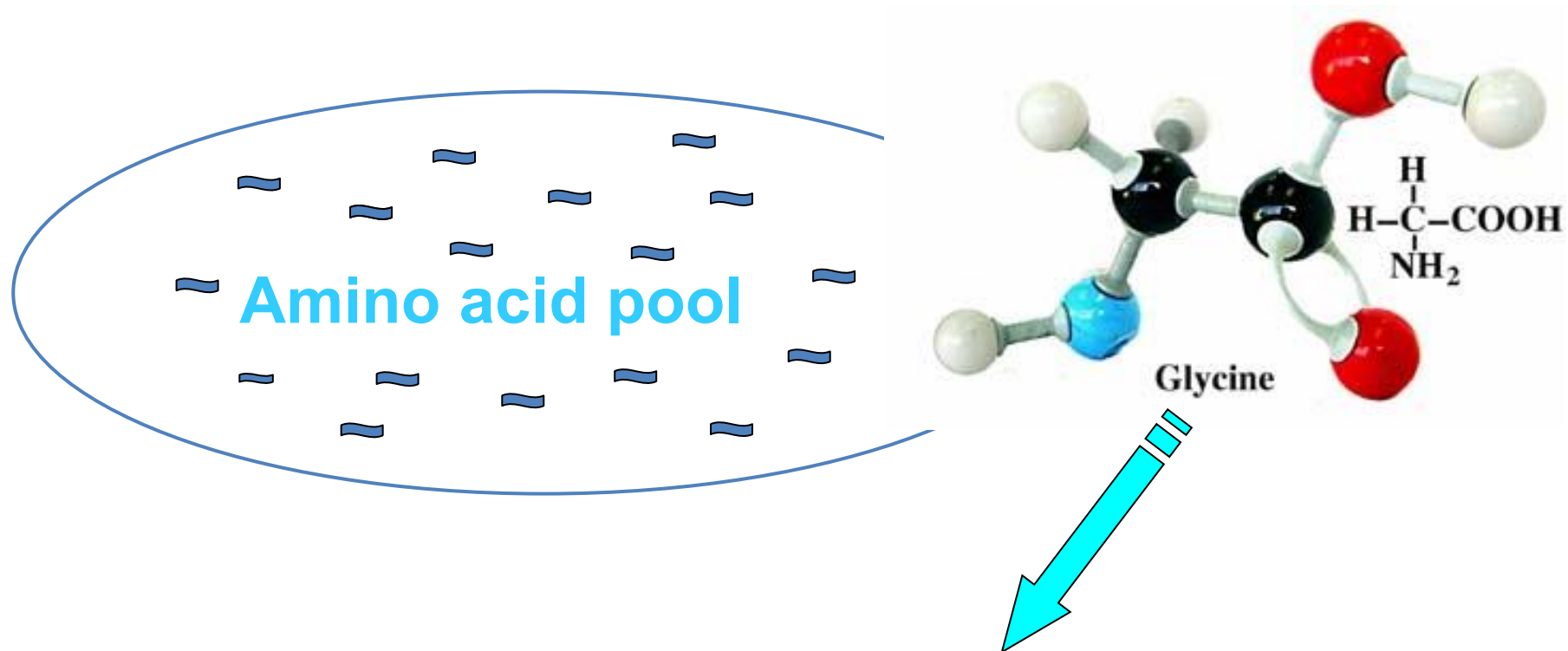
- Lippincott Illustrated Reviews (**Lippincott**) 9th edition; Chapter 21: Pages 308 – 313; Section: Porphyrin metabolism; [Amino Acids: Conversion to Specialized Products | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Lippincott's Biochemistry 9th Edition
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Mark's Basic Medical Biochemistry: A clinical approach (**MBMB**) 6th edition; [The Biochemistry of Erythrocytes and Other Blood Cells | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library](#)
- www.haematologica.org/content/99/973 Heme and erythropoiesis: more than a structural role. Review article D. Chiabrando et al, Hematologica 2014
- <http://porphyria.eu/en/content/regulation> **ALAS1** -glucose-and-heme
S Kolluri et al. European Porphyria Network 2018

Objectives: Heme Synthesis and Porphyrrias

SOM. MK.I.BPM2.3.DM.2.BCHM.1267	Describe significance of heme synthesis. Outline concept of heme synthesis starting from glycine and succinyl CoA to the formation of heme.
SOM. MK.I.BPM2.3.DM.2.BCHM.1268	Distinguish between regulation of heme synthesis in liver and in erythroid cells.
SOM. MK.I.BPM2.3.DM.2.BCHM.1269	Describe reaction catalyzed by ALA synthase and explain how pyridoxine deficiency affects heme synthesis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1270	Describe reaction catalyzed by ALA dehydratase and discuss effects of lead poisoning on heme synthesis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1271	Classify porphyrias based on clinical manifestations such as photosensitivity, abdominal pain and neuropsychiatric manifestations.
SOM. MK.I.BPM2.3.DM.2.BCHM.1272	Describe metabolic basis for clinical features occurring in acute intermittent porphyria (AIP). Identify the deficient enzyme and products accumulating in blood and urine.
SOM. MK.I.BPM2.3.DM.2.BCHM.1273	Discuss the molecular basis for use of hemin in management of AIP.
SOM. MK.I.BPM2.3.DM.2.BCHM.1274	Explain the clinical significance of avoiding phenobarbital in patients with acute intermittent porphyria (AIP)
SOM. MK.I.BPM2.3.DM.2.BCHM.1275	Describe the metabolic basis for clinical features and laboratory findings in congenital erythropoietic porphyria. Identify the deficient enzyme and products accumulating in blood and urine.
SOM. MK.I.BPM2.3.DM.2.BCHM.1276	Describe the metabolic basis for clinical features of porphyria cutanea tarda. Identify the deficient enzyme and products accumulating in blood and urine.

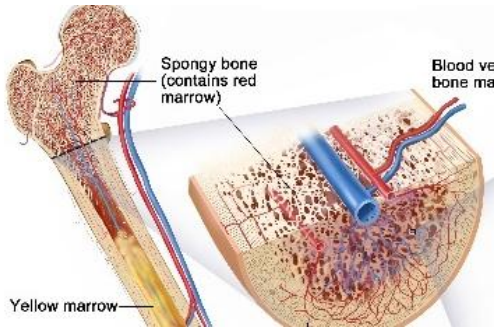
Conversion of amino acids to specialized products

Heme synthesis



Synthesis of heme: **glycine** is used

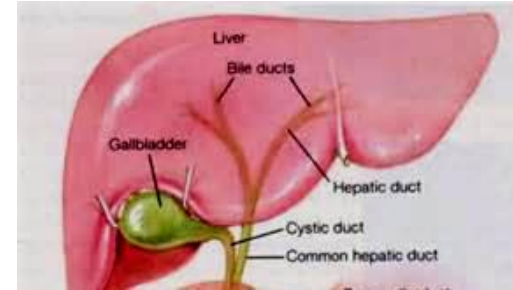
Sites of heme synthesis



Bone marrow:

Hemoglobin synthesis

- Synthesis in erythroblast and reticulocytes
- **Iron availability** regulates heme synthesis
- Heme stimulates globin chain synthesis for hemoglobin formation



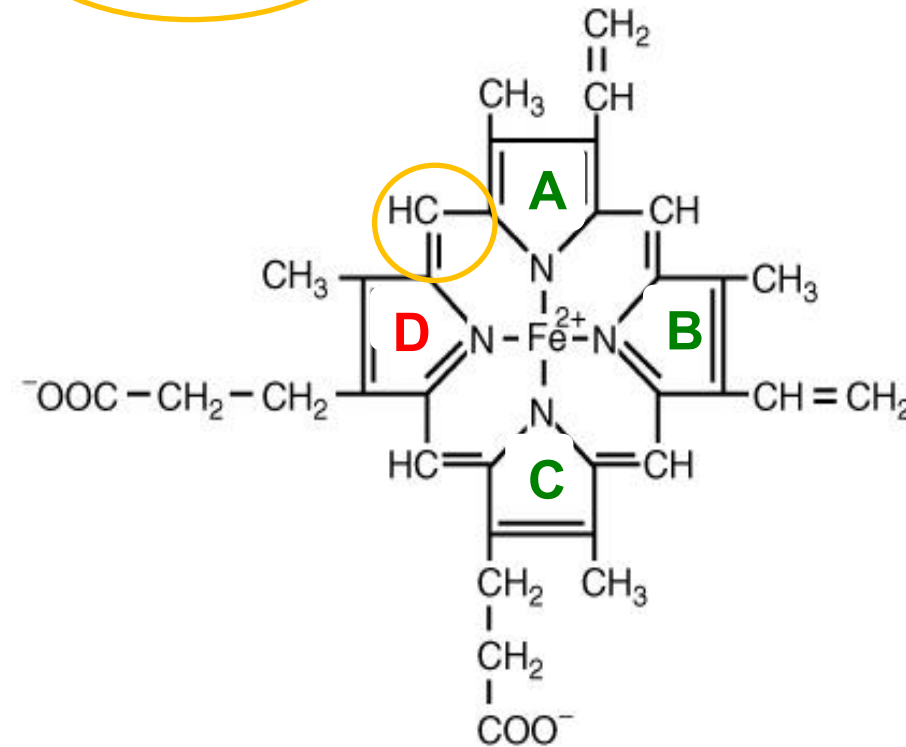
Liver:

Variable CYP450 synthesis

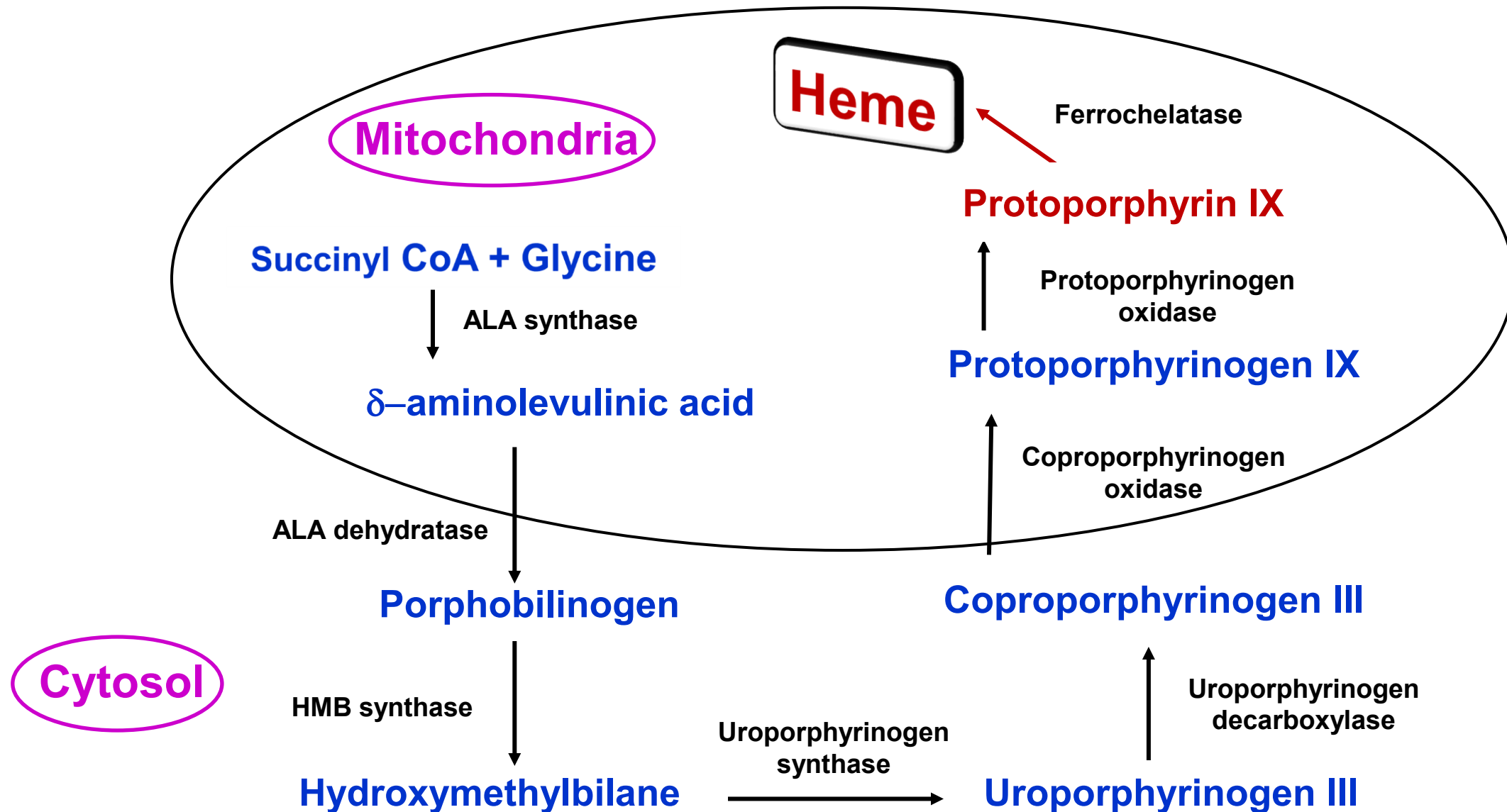
- Heme is needed as prosthetic group for enzymes and cytochromes
- [Heme] regulates heme synthesis
- Excess Heme does not accumulate

Heme (ferrous protoporphyrin IX)

- 4 pyrrole rings (A-D) are linked via methenyl bridges.
- Side chains of heme:
 - **A**: methyl, vinyl
 - **B**: methyl, vinyl
 - **C**: methyl, propionyl
 - **D**: propionyl, methyl
- Ring D is asymmetric



Overall Big Picture of Heme synthesis



Substrates

Succinyl CoA + Glycine

δ-aminolevulinic acid (**ALA**)

Porphobilinogen
(**PBG** from 2 **ALA**)

Hydroxymethylbilane
(**HMB** from 4 **PBG**)

Heme

ferrous iron

Protoporphyrin IX (red)

Protoporphyrinogen IX

Coproporphyrinogen III

Uroporphyrinogen III

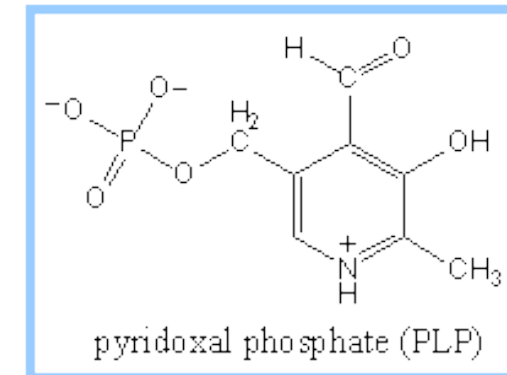
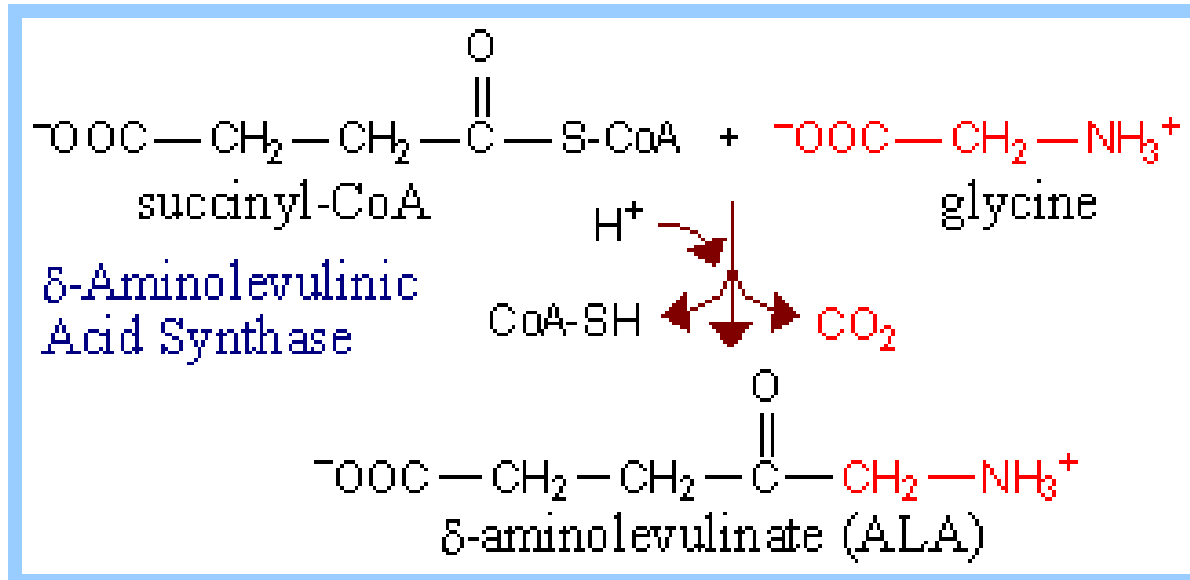
Text: Heme synthesis in all cells

- Use the same building blocks. Start and end in mitochondria. All reactions are irreversible

1. **Succinyl CoA and glycine** → δ -aminolevulinic acid (ALA) (mitochondria)
2. **ALA** leaves mitochondria and 2 ALAs form **PBG** (cytosol)
3. **Four porphobilinogens** are aligned and deaminated to the **linear HMB** (cytosol)
4. **Enzymatic ring closure of HMB** forms **Uroporphyrinogen III** (cytosol)
5. Decarboxylation of uroporphyrinogen III to **Coproporphyrinogen III** (cytosol)
6. **Coproporphyrinogen III** transported into mitochondria
7. Changes in Double-bonds to form red **protoporphyrin IX** (mitochondria)
8. **Protoporphyrin IX** + ferrous iron (Fe^{2+}) → **Heme** (mitochondria)

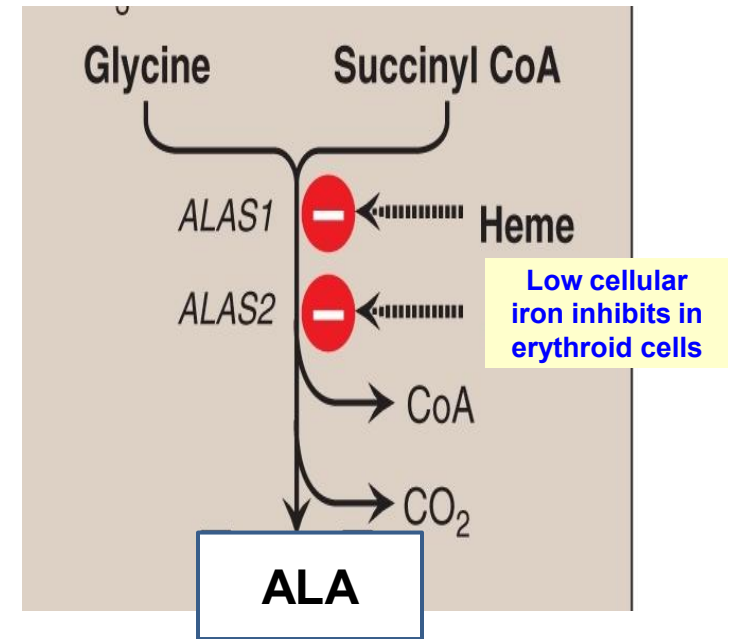
Heme synthesis: Regulated Step

- Succinyl CoA + Glycine → ALA (decarboxylation)
- Enzyme: **ALA synthase**
- Coenzyme: **Pyridoxal phosphate (PLP)** from **vitamin B6**

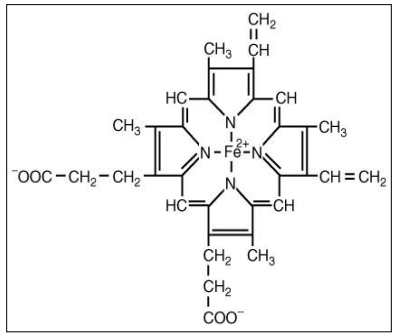


Regulation of ALA synthase

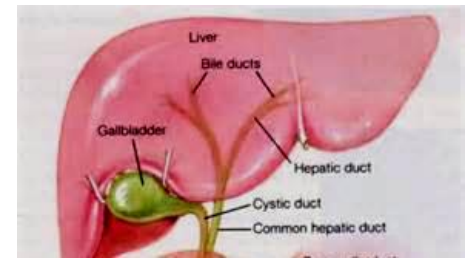
- Different regulation in liver vs. bone marrow
- Liver (& most cells) → ALAS1
 - **low heme** concentration → **↑ Gene expression**
 - **high heme** concentration → feedback **inhibition**
- Bone marrow (Erythrocyte precursors): ALAS2
 - **high** intracellular **iron** → **stimulates** translation of ALAS2 mRNA
 - **low** intracellular **iron** → **inhibits** ALAS2 mRNA translation



Modified L p. 278 7th edition



Liver: Heme synthesis



Heme synthesis in liver is regulated by free heme:

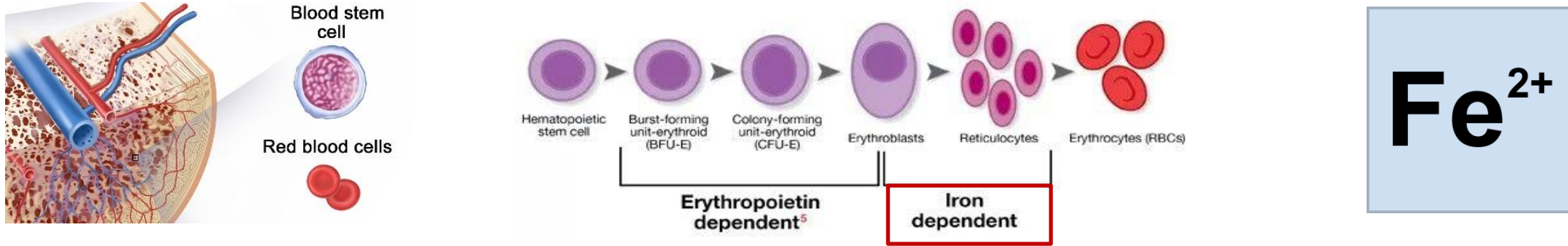
1. Inhibition of ALAS1 synthesis:

- ↑ levels of heme reduces ALAS1 expression
- **Hemin** (drug) used in acute intermittent porphyria

2. Stimulation of ALAS1 synthesis:

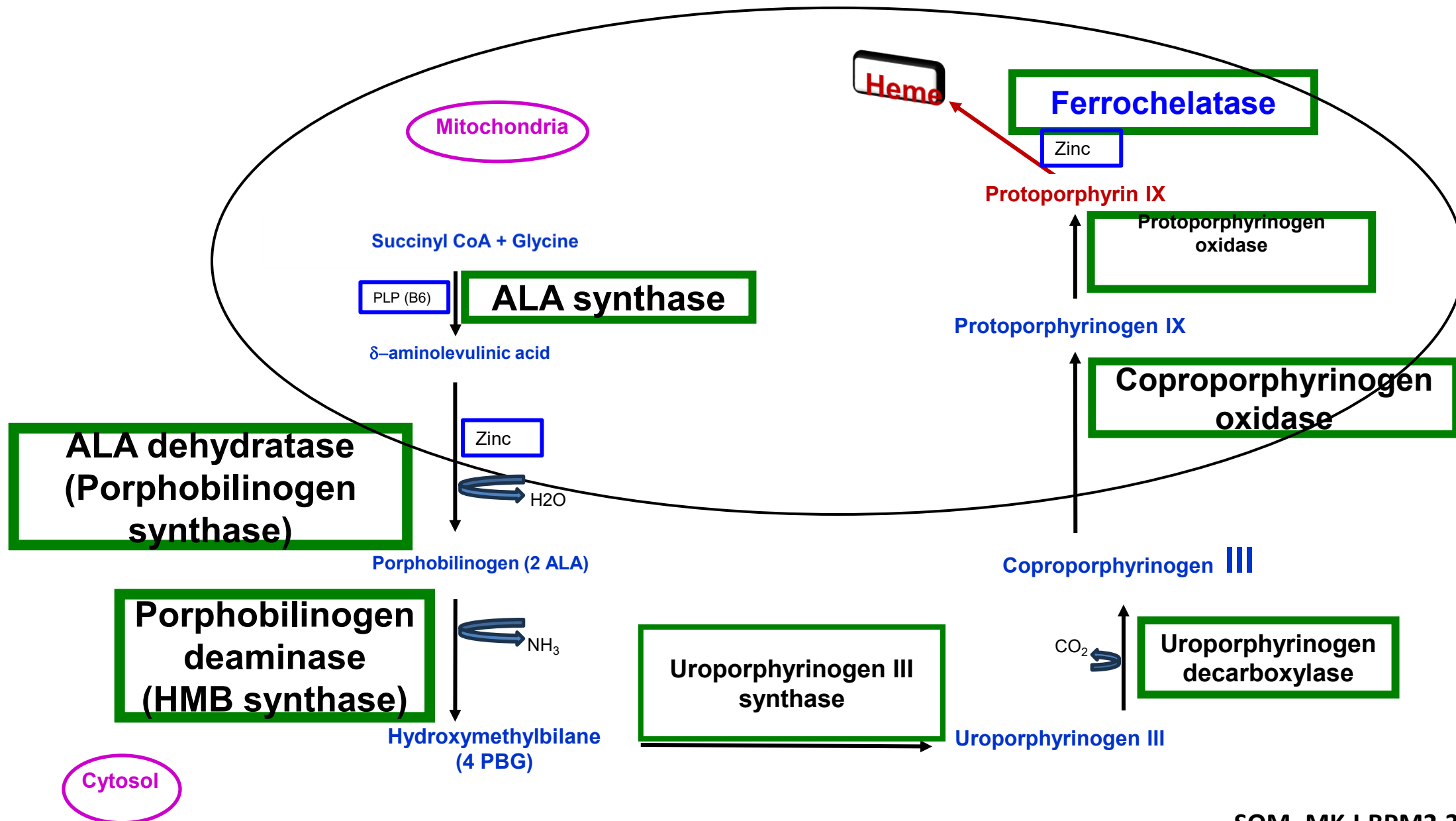
- ↓ intracellular heme concentration stimulates transcription.
- Certain drugs (**barbiturates**) induce CYP 450 increase ALAS1 synthesis.
- **Avoid barbiturates** in acute intermittent porphyria

Bone marrow: Hemoglobin synthesis



- **Hypoxia** stimulates **Erythropoietin** (kidney) → RBC and hemoglobin synthesis.
- Iron uptake is stimulated (in erythroblasts) → Increased heme synthesis
- **ALAS2 synthesis**
 - ✓ regulated by **intracellular iron** (not inhibited by heme)
 - ✓ **mRNA** of ALAS2 has an **iron-response element (IRE)**
 - ✓ translated only when iron is present
- **Free heme** increases expression of **hemoglobin α -** and **β -globin genes** (ensures balanced synthesis of hemoglobin)
- **ALAS2 gene (X-chromosome):** Deficiency leads to **X-linked sideroblastic anemia**.

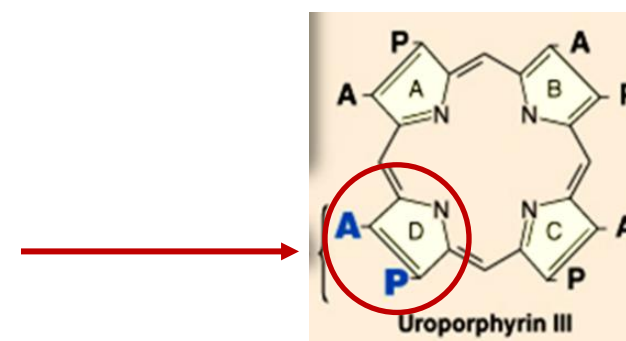
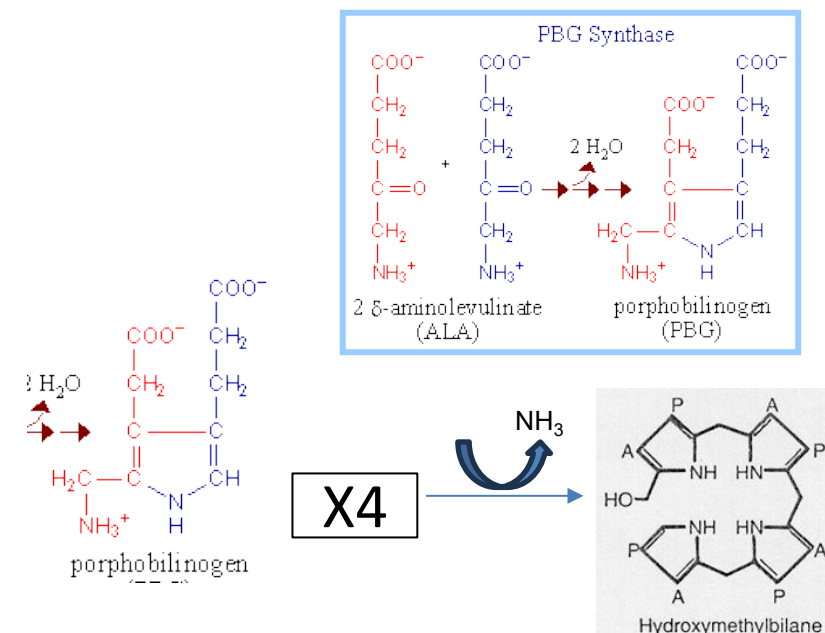
Enzymes of Heme synthesis



Functions of ALA dehydratase and PBG deaminase

1. **ALA dehydratase (PBG synthase)** uses **2 ALAs** to form **porphobilinogen**. **Zinc** cofactor.
2. **PBG deaminase** aligns and **deaminates 4 porphobilinogens** to form **linear hydroxymethylbilane (HMB)**.
3. Enzymatic **ring closure** of **HMB** forms the **porphyrin III** ring system and leads to an **asymmetric pyrrole ring D**.

www.rpi.edu/dept/bcbp/molbiochem/MBWeb/mb2/part1/heme.htm



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Inhibition of heme synthesis & **microcytic anemia**

1. **Deficiency of vitamin B₆**

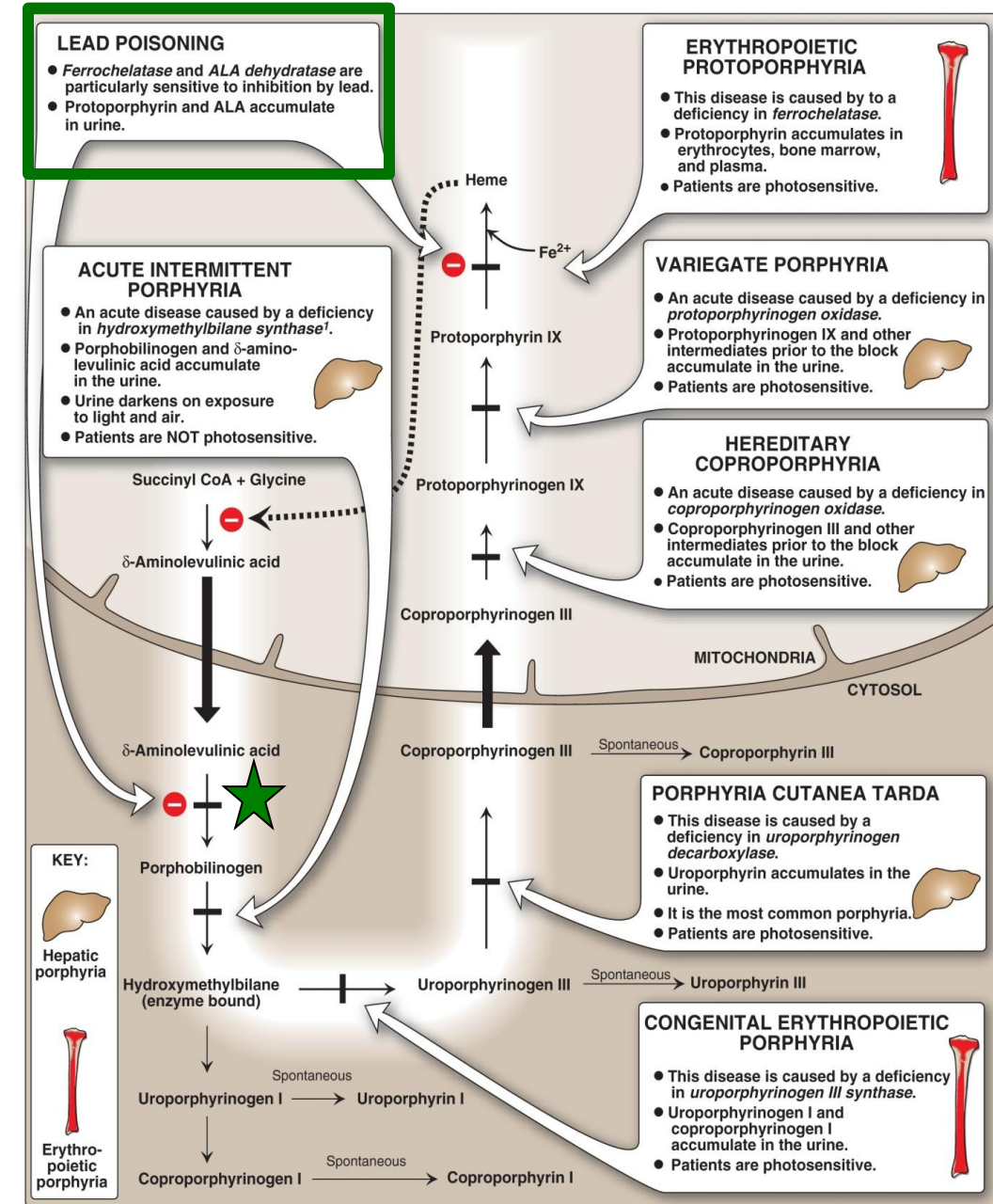
- **ALA synthase** needs **pyridoxal phosphate (PLP)** as **coenzyme**
- **↓ pyridoxal phosphate → ↓ Heme synthesis** in all cells
- **Isoniazid** (tuberculosis treatment) → **PLP depletion**
- **Treatment: Vitamin B6 supplementation**

2. **Lead poisoning**

- **ALA dehydratase** and **ferrochelatase** need **zinc** as cofactor.
- **Lead (heavy metal) interacts with zinc → ↓ heme synthesis**
- **ALA and protoporphyrin IX accumulate in blood and urine.**

Disorders of porphyrin metabolism

- **Lead poisoning:** Inhibits **Ferrochelatase** and **ALA dehydratase** : $\uparrow$ ALA and protoporphyrin IX in blood and urine
- **Porphyrias:**
 - **Acute Intermittent Porphyria (AIP)**
 - **Congenital Erythropoietic Porphyria (CEP)**
 - **Porphyria Cutanea Tarda (PCT)**
- Note: Enzyme deficiencies after HMB lead to skin-photosensitivity due to excess abnormal porphyrins



Porphyrias

1. Enzyme deficiency (**genetic or acquired**)

All porphyrias are **autosomal dominant** traits (**except CEP**)

2. **Accumulation of intermediates of heme synthesis** in blood, various tissues and urine.

3. **Clinical features:**

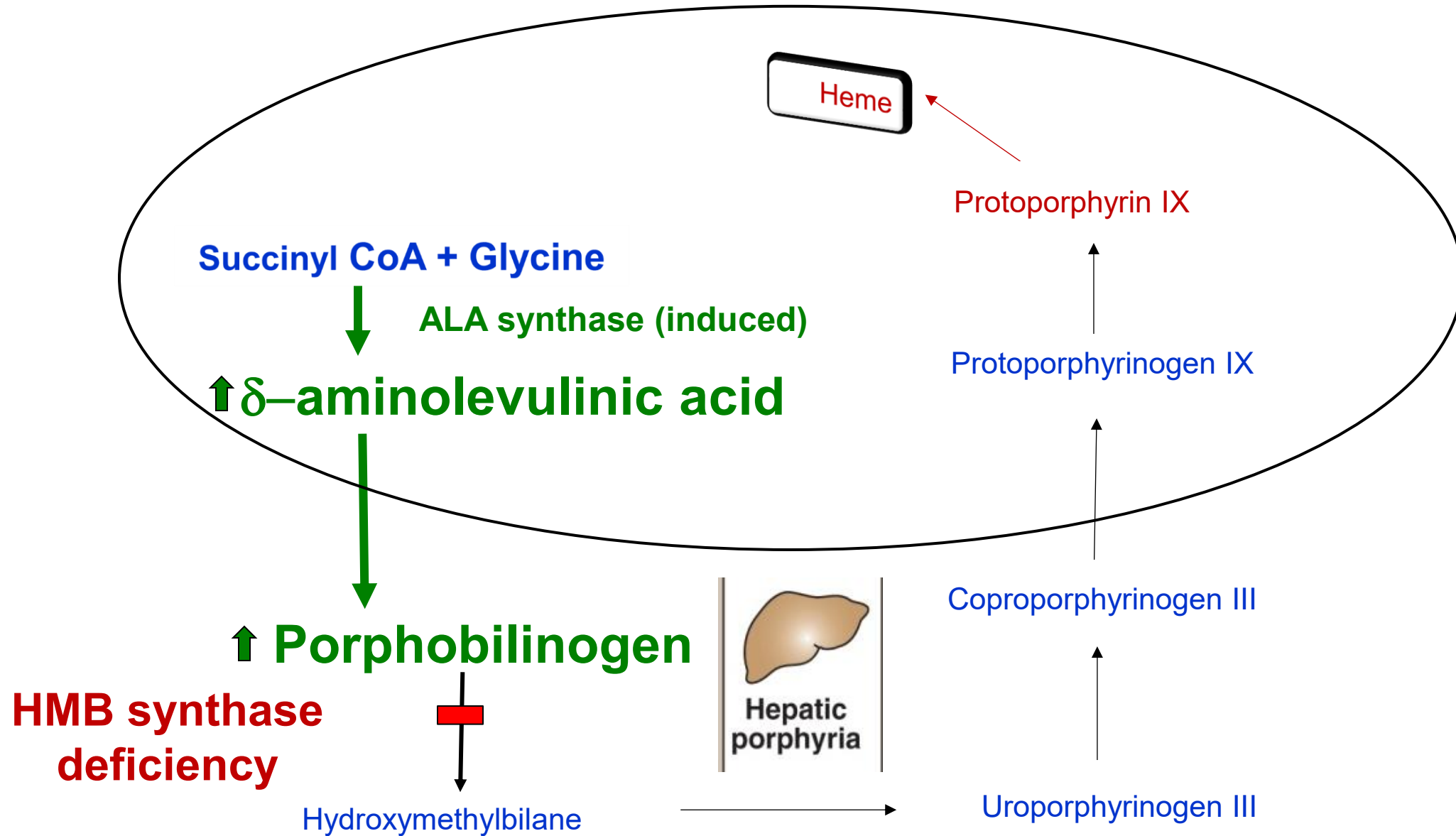
- a. Severe abdominal pain, neurologic dysfunction and mental disturbance (**AIP**)
- b. Photosensitivity of skin (**PCT** and very severe in **CEP**)

4. Grouped into **hepatic** or **erythropoietic porphyrias**

5. **Hepatic porphyrias:**

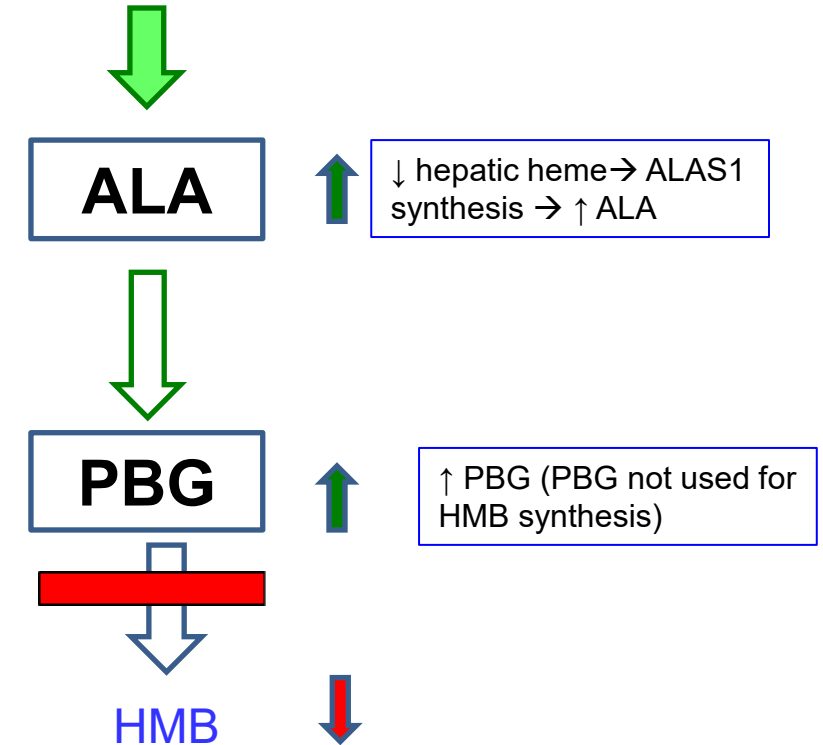
- **Treatment with intravenous glucose and saline:** A metabolite of glucose inhibits ALAS1; Hypoglycemia induces heme synthesis
- **Treat with Hemin infusion:** contains ferric iron and inhibits ALAS1
- Symptoms worsened by **drugs** that induce hepatic **Cytochrome P450 (barbiturates)**

HMB synthase deficiency : Acute Intermittent Porphyria



Acute Intermittent Porphyria (AIP)

- **Severe deficiency of PBG deaminase (HMB synthase)**
- Many mutations of PBG deaminase gene
- Up-regulation of hepatic ALAS1 → no feedback inhibition by heme



Acute intermittent porphyria: Clinical features

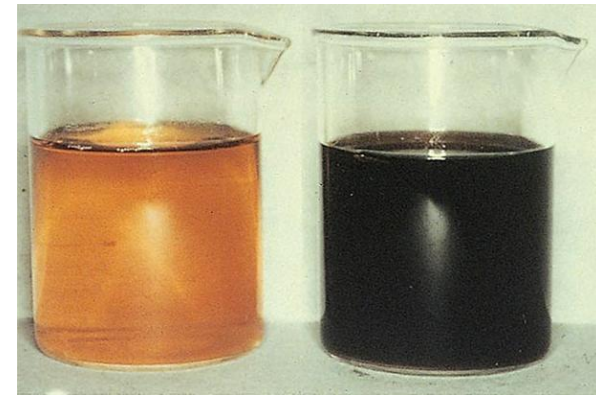
Clinical features:

- Visceral:
 - Episodic severe abdominal pain; Abdominal colic
 - Constipation
 - **ALA (structure similar to GABA)** and competes for **GABA receptor**
- Neurologic:
 - Weakness of lower extremities
 - Hypertension, Tachycardia and respiratory problems
 - **Highly agitated state and mental disturbance**

NOTE: NO **photosensitivity** (HMB and abnormal porphyrins not formed)

Laboratory tests:

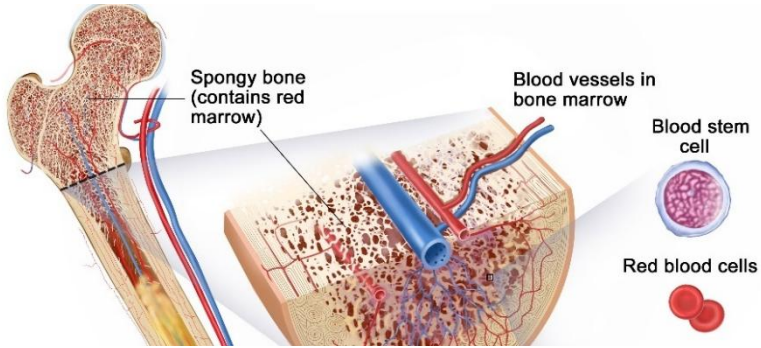
- **Urinary ALA and porphobilinogen:** ↑↑
- **Urine analysis:** Rapid dipstick method for **porphobilinogen**.
- **Urine color change** (exposure to air/light): Porphobilinogen (colorless)
→ **red purple-black porphobilin**



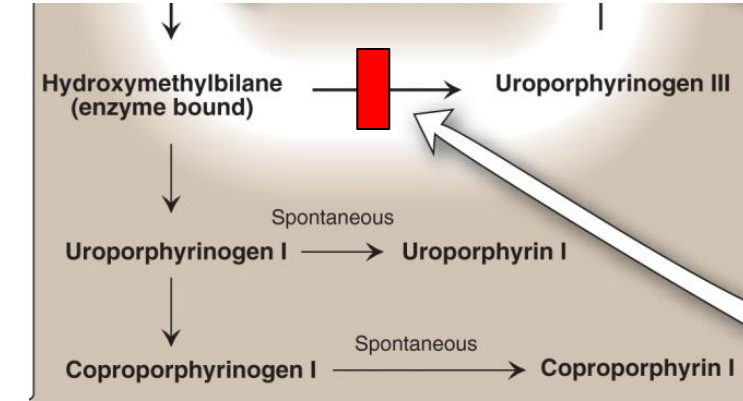
AIP: Summary

- Onset
 - More common **Caucasians**
 - **Predominantly women** (2nd to 4th decade)
- Triggers: Inducers of hepatic ALAS1 and hepatic cytochrome P450 enzymes:
 - Drugs: (**Barbiturates**, phenytoin, other anticonvulsants); Hormone therapy (progesterone)
 - Ethanol; Smoking
 - Infections
- Treatment: immediate for **pain** and **vomiting**
 - **Saline; IV glucose**
 - **hemin**
 - Note: A **glucose metabolite** and **hemin** → ↓ **ALAS1** gene expression; and **hypoglycemia** → ↑ **ALAS1**
- Avoid **barbiturates** or **drugs** that stimulate **cytochrome P450 synthesis**.
 - applies to **treatment** of **hepatic porphyrias**
 - **Barbiturates** lead to **toxic ALA levels** in AIP patients and can lead to **death**
- **Misdiagnosis and delay in diagnosis is common**

Congenital Erythropoietic Porphyria (CEP)

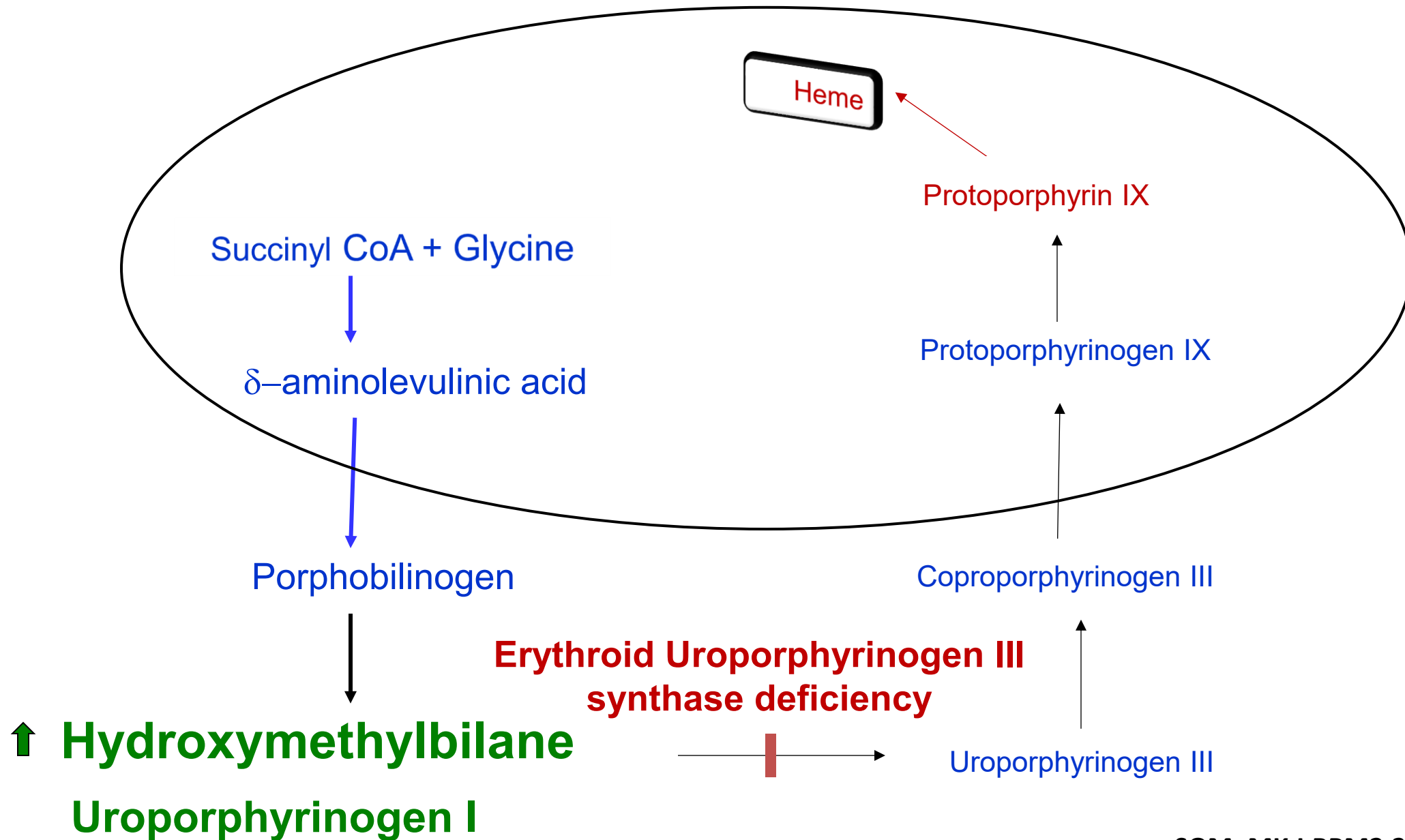


Heme synthesis

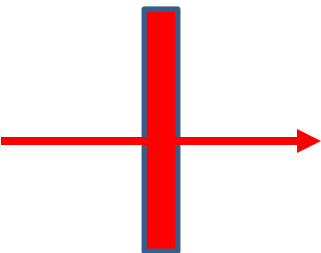
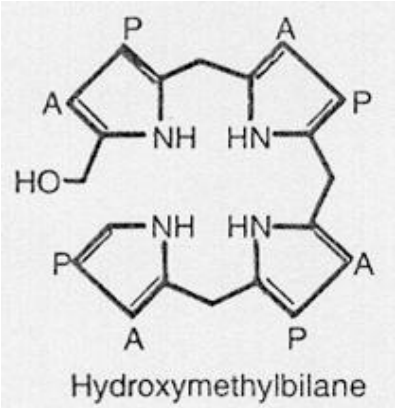


- Hereditary defect only in **bone marrow**
- Autosomal recessive disorder
- Deficiency: **Uroporphyrinogen III synthase (erythroid isozyme)**
- Leads to **abnormal porphyrins** and severe **skin photosensitivity**

CEP: Uroporphyrinogen III synthase deficiency (bone marrow)



CEP: Accumulation of type I porphyrins



Deficiency of erythroid
uroporphyrinogen III synthase

HMB spontaneously forms
abnormal type I porphyrin



Uroporphyrinogen I



Coproporphyrinogen I



Uroporphyrin I



Coproporphyrin I



CEP: Formation of Uroporphyrin-I and coproporphyrin-I

- Enzymatic **Ring closure** of **HMB** generates **uroporphyrinogen III**.
- **Erythroid uroporphyrinogen III synthase** deficiency (CEP) → HMB (linear) spontaneously forms **type I abnormal porphyrin ring**
- Uroporphyrinogen I is then decarboxylated to coproporphyrinogen I.
- **Uroporphyrinogen I** and coproporphyrinogen I → **uroporphyrin I** and **coproporphyrin I (red)**
- Accumulates in **blood, skin** and **tissue** → **ROS** formation in **skin** → **severe, painful photosensitivity** (abnormal porphyrins are not added to proteins)
- The **urine** is **red** upon **release**.

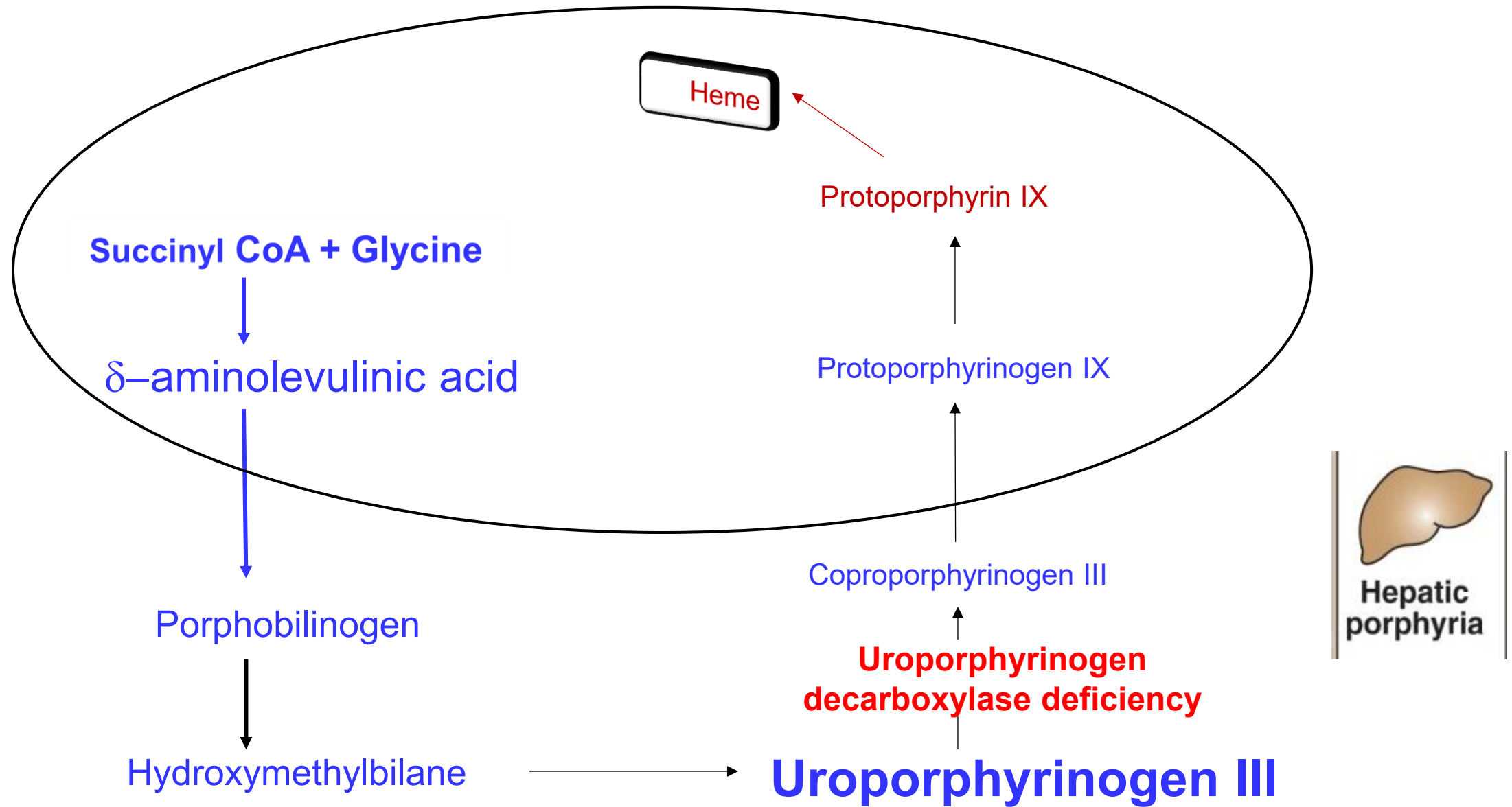
CEP: Clinical features and management

- Onset: mainly in **childhood** or earlier
- Severe, painful skin blisters and ulcers (UV light exposure)
- Poor wound healing
- **Teeth: reddish-brown**
- **Hypertrichosis:** Werewolf features, hairy face and arms
- **Treatment:** Bone marrow transplantation
- **Note:** Hemin not effective → ALAS2 gene expression not inhibited by heme

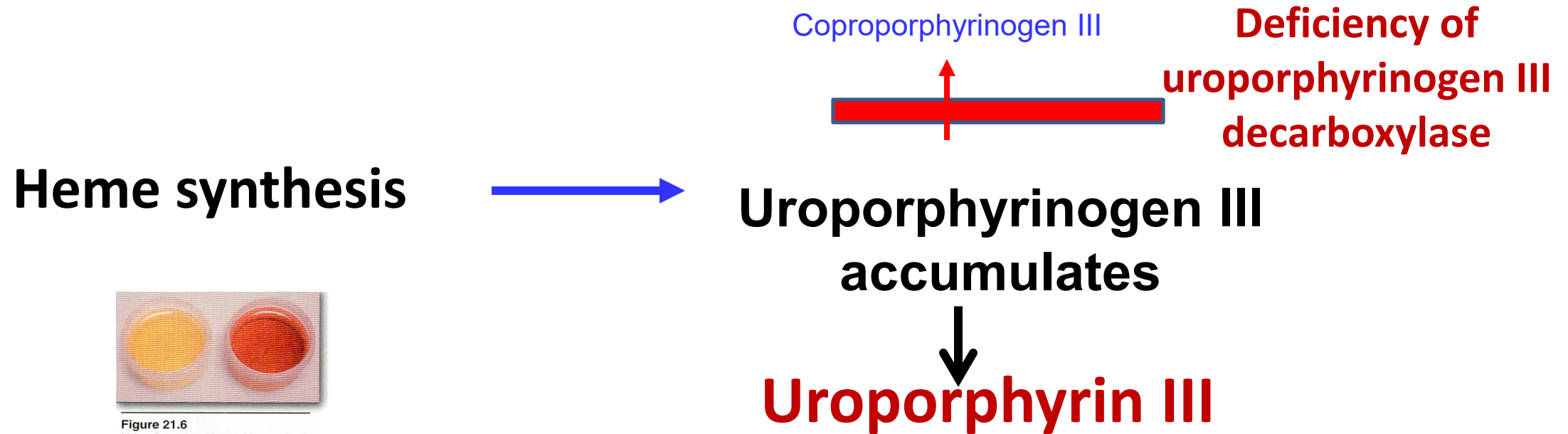


http://www.gfmer.ch/genetic_diseases-v2/gendis_detail_list.php?cat3+118

Uroporphyrinogen decarboxylase deficiency in liver (PCT)



Porphyria Cutanea Tarda (PCT)



- ↑ **Uroporphyrin III** (liver, blood, skin) → cutaneous lesions (photosensitivity)
- **Urine: red** upon release in natural light or **pink** in fluorescent light

PCT: Onset and treatment

- Onset: **40-50 years** of age
- Classification
 - **PCT Type I** (Acquired: 80%)- **Chronic liver disease** (hepatitis or ethanol abuse)
 - **PCT Type II**- **familial autosomal dominant**
- Disease **expression** influenced by:
 - Hepatic iron overload
 - Exposure to sunlight
 - Excessive alcohol intake
- **Management**
 - **Avoid** sunlight and alcohol
 - Reduce dietary iron intake



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Summary: Disorders **not** leading to photosensitivity

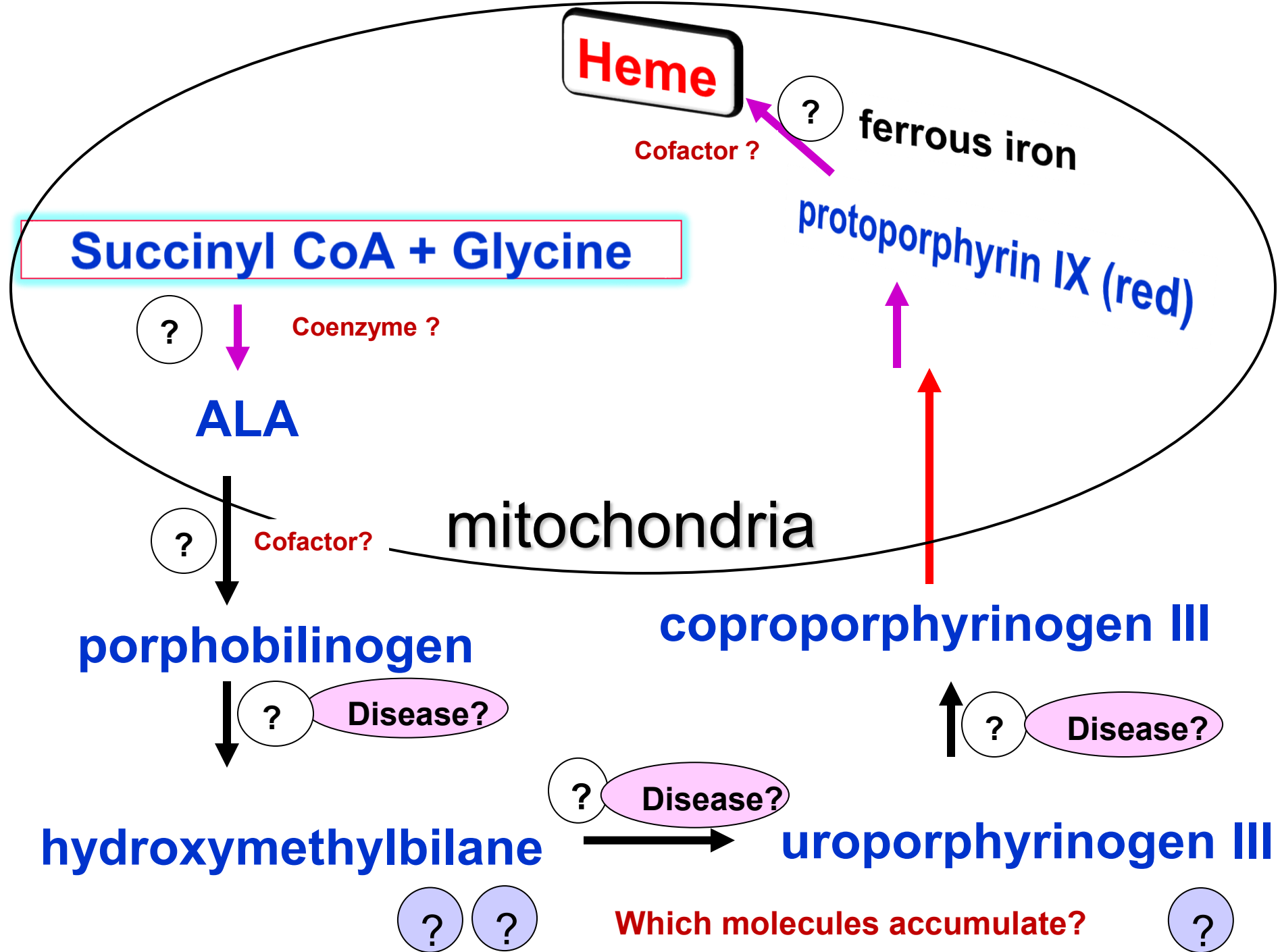
Vitamin B6 deficiency	• Deficiency of ALA synthase which needs PLP synthesized from B6. • Can be caused by dietary deficiency but also by isoniazid treatment.
Lead poisoning	Deficiency of ALA dehydratase and ferrochelatase Accumulation of ALA and protoporphyrin IX in blood and urine
Acute Intermittent Porphyria (triggered)	Deficiency of PBG deaminase (HMB synthase) Accumulation of ALA and porphobilinogen in blood and urine

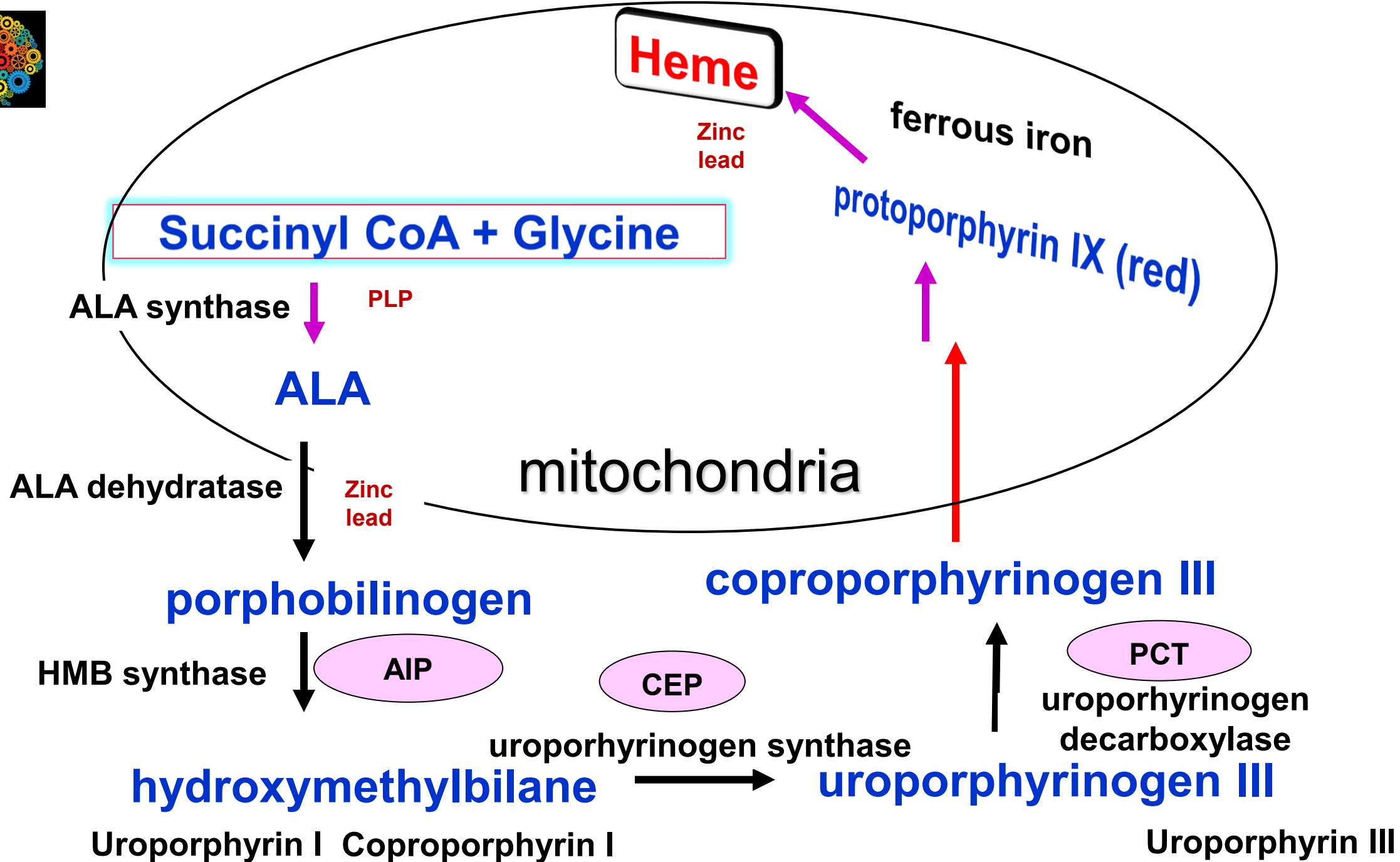
Summary: Disorders with Skin photosensitivity

Concept: Defects of heme synthesis after formation of HMB lead to skin photosensitivity of patients due to abnormal porphyrins. The skin itches and burns when it is exposed to light (ROS formation) due to abnormal porphyrins which are not incorporated into proteins .	
Congenital Erythropoietic Porphyria (CEP)	• Deficiency of <u>erythroid</u> uroporphyrinogen III synthase. • Most severe photosensitivity. • Blood, tissue, skin and urine show accumulation of uroporphyrin I & coproporphyrin I.
Porphyria Cutanea Tarda (PCT)	• Deficiency of uroporphyrinogen decarboxylase. • Accumulation of uroporphyrin III in blood, tissue, skin and urine.

Worksheet suggestion	 Acute Intermittent Porphyrria (AIP)	Congenital Erythropoietic Porphyrria (CEP)	Porphyrria Cutanea Tarda (PCT)
Deficient enzyme			
Cells			
Intermediates accumulated			
Clinical features			
Skin photosensitivity			
Explain.			
Color of urine upon release and on standing			
Treatments			
Typical age of onset			

Worksheet suggestion 	Acute Intermittent Porphyria (AIP)	Congenital Erythropoietic Porphyria (CEP)	Porphyria Cutanea Tarda (PCT)
Deficient enzyme Which cells	Porphobilinogen deaminase (HMB synthase) in hepatocytes	Uroporphyrinogen III synthase in erythroid cell of bone marrow only	Uroporphyrinogen decarboxylase in hepatocytes
Intermediates accumulated Clinical features	ALA and porphobilinogen Triggered, highly agitated, pain.	Uroporphyrin I and coproporphyrin I Severe blisters	Uroporphyrin III Triggered by sunshine or alcohol, blisters
Skin photosensitivity Explain	No, HMB is not formed, no abnormal porphyrins are formed	Yes, abnormal red porphyrins Type I	Yes, abnormal red uroporphyrin III
Color of urine upon release and on standing	Normal color upon release, darkens upon standing due to porphobilin formation	Red	Red, pink fluorescent light
Treatments	Pain treatment and infusion with glucose or hemin. NO admin. of barbiturates!	Pain treatment and protection against sun. Bone marrow transpl.	Protection against sun and no ingestion of alcohol. Low iron diet
Typical age of onset?	Adult, onset is triggered	Infant and childhood	Adult





**Thank you, if you have questions,
please email.**

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